

Charge-Quantum Distribution and Topological Configuration in Leptons

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Abstract

We present a geometric model for the charge distribution and topological configuration of leptons based on primitive charge quanta. The electron e^- is constructed as a five-body two-dimensional square cell consisting of four peripheral negative charge quanta $\bar{\xi}$ with charge $q(\bar{\xi}) = -e/3$ located at the square vertices and a central positive charge quantum ξ with charge $q(\xi) = +e/3$ positioned at the centroid, yielding the exact net electric charge $Q_e = -e$. The formulation proceeds in two stages. In the first stage, a static geometric configuration is established through the pair potential $V(r) = -k/r + A/r^2$, leading to force-balance equilibrium, an equilibrium cell radius $R_e = 0.4627$ fm, and a finite static binding energy, with the interaction constant calibrated from the up-quark mass ($m_u = 2.16$ MeV). The finite equilibrium radius naturally regularizes the classical short-distance Coulomb singularity, yielding finite internal potential and self-energy while preserving the standard $1/r$ Coulomb interaction outside the electron core. In the second stage, trapped kinetic energy is interpreted as localized circulating vacuum excitations about the equilibrium sites. Under the double-cover mapping $D_4 \rightarrow SU(2)$, the circulating configuration recovers intrinsic spin $S = \hbar/2$ and the tree-level Dirac magnetic moment $\mu_e = -\mu_B$, while the electromagnetic energy of the external monopole field, truncated at the cell boundary, yields the electron rest energy $m_e c^2 = \frac{3k}{2R_e} = 0.5187$ MeV, within approximately 1.5% of the observed value. The framework is further extended to the electron neutrino, modeled as an alternating ring composed of two charge-quantum/anti-charge pairs, predicting a heavy Majorana state with radius $R_\nu = 0.02171$ fm and mass $M_\nu = 20.21$ MeV. Geometric scaling of the cell metric gives a Dirac mass $m_D = m_e^{\text{obs}} \left(\frac{R_\nu}{R_e} \right)^2 = 1.125$ keV, which, through the Seesaw mechanism, produces a light neutrino mass $m_\nu^{\text{light}} = \frac{m_D^2}{M_\nu} = 0.0626$ eV, consistent with the atmospheric neutrino oscillation scale without introducing additional free parameters.

1 Introduction

Quantum electrodynamics describes the electron as a point-like Dirac field. This description has been verified with extraordinary precision and forms one of the cornerstones of modern particle physics. In perturbative calculations, the point-like description requires the renormalization procedure to obtain finite observable quantities. Renormalization provides a mathematically complete and experimentally successful framework for calculating physical predictions. The present work explores whether the experimentally determined intrinsic properties of the electron may also admit an underlying geometric and topological description.

Neutrino physics provides another important motivation for exploring possible underlying structures. The observation of neutrino oscillations [7] establishes that neutrinos possess non-zero masses, whose values are many orders of magnitude smaller than those of the charged leptons. Extension models, including the seesaw mechanism [2, 3, 4], provide elegant frameworks for generating naturally small neutrino masses. The origin of the corresponding mass scales, however, remains an active subject of theoretical investigation.

These observations motivate the exploration of whether the empirical parameters admit a deeper geometric or topological interpretation. Just as the structure of atoms provided the foundation for understanding the periodic table, it is natural to ask whether the observed properties of elementary particles originate from an underlying organization of more primitive vacuum structures.

In this paper, we investigate one possible realization based on the hypothesis of charge quantisation and pair production [1]. In this work electric charges are assumed to be generated as a pair of primitive charge quanta, $(\xi, \bar{\xi})$, carrying fractional charges $\pm e/3$ of opposite sign. The up and down quarks are represented as stable topological assemblies of these charge quanta, with their physical properties determined by the geometry, symmetry, and vacuum-energy distribution of the corresponding configurations.

The primary objectives of this work are:

- i. To establish the minimal stable charge-quantum topologies for the electron and the neutrino.
- ii. To derive the electron radius, rest mass, intrinsic spin, and magnetic moment from the proposed geometric construction without introducing adjustable parameters.
- iii. To derive the neutrino radius, the heavy Majorana mass, and the light neutrino mass through the same geometric framework together with the seesaw mechanism.
- iv. To investigate whether these geometric constructions provide a structural interpreta-

tion of particle properties that complements the conventional understanding.

The paper is organized as follows. Section 2 reviews the charge-quantization theorem together with the charge-quantum assignments and topologies of quarks and leptons, establishing the foundation of the model. Section 3 develops the static geometric description of the electron, deriving its equilibrium radius and binding energy from force balance. Section 4 extends the construction to the relativistic field, deriving the intrinsic spin, magnetic moment, and rest mass. Section 5 applies the same principles to the neutrino ring, obtaining its radius together with the heavy and light neutrino masses. Section 6 concludes with a summary and discussion of the implications.

The principal results obtained in this work are summarized below:

- i. The electron is described by a five-charge-quantum square topology $(4\bar{\xi} + \xi_0)$ with equilibrium radius $R_e = 0.4627$ fm obtained from force balance.
- ii. The electron rest mass is derived as $m_e c^2 = 3k/(2R_e) = 0.5187$ MeV, within approximately 1.5% of the measured value.
- iii. The electron spin $S = \hbar/2$ and magnetic moment $\mu_e = -\mu_B$ emerge from the $D_4 \rightarrow SU(2)$ double-cover symmetry of the square topology.
- iv. The neutrino is represented by an unanchored four-segment alternating ring with radius $R_\nu = 0.02171$ fm and heavy Majorana mass $M_\nu = 20.21$ MeV.
- v. Using the geometric Dirac mass $m_D = m_e^{\text{obs}}(R_\nu/R_e)^2 = 1.125$ keV together with the seesaw mechanism yields a light neutrino mass $m_\nu^{\text{light}} = 0.0626$ eV, consistent with the atmospheric neutrino oscillation scale.

Within the proposed geometric model, the finite electron radius leads to a finite classical electromagnetic self-energy. More generally, the framework investigates whether quantities that appear as empirical inputs may admit an underlying geometric interpretation in terms of charge-quantum topology.

We emphasize that the present work is not intended to replace quantum field theory, but to explore a complementary geometric layer for the structural interpretation of particle properties. If such a description proves physically relevant, it would provide an underlying geometric perspective consistent with the successful effective field-theoretic description.

2 Charge Quantisation and Charge-Quantum Distribution in Fermions

This work builds directly upon the charge quantisation theorem established previously [1]. The essential results of the model developed in [1], particularly required for the present development are briefly summarized here.

2.1 Charge Quantisation and Pair Production

The charge quantisation theorem identifies two elementary vacuum charge quanta,

$$\xi \equiv +\frac{e}{3}, \quad \bar{\xi} \equiv -\frac{e}{3}, \quad (1)$$

from which all observed electric charges are constructed as integer combinations. The theorem further establishes that charge quanta are never created or annihilated individually, but instead the creation and annihilation always occur through complementary pairs,

$$(\xi, \bar{\xi}), \quad (2)$$

thereby preserving exact global charge neutrality,

$$\xi + \bar{\xi} = 0. \quad (3)$$

The production aspect of this theorem was demonstrated previously through radioactive beta decay, while the annihilation aspect was established through electron–positron annihilation. Consequently, every charged particle must ultimately be composed of these elementary charge quanta. Within the proposed model, the electric charge of every fermion is obtained by summing the charges of its constituent charge quanta,

$$Q = \sum_i q_i. \quad (4)$$

2.2 Charge Quantum Distribution and Topologies in Quarks

In the proposed model, quarks are composed of the primitive vacuum charge quanta ξ and $\bar{\xi}$ as defined in Section 2.1. The up and down quarks are represented by the configurations

$$u \equiv (\xi, \xi, \xi, \bar{\xi}), \quad d \equiv (\bar{\xi}, \xi, \bar{\xi}). \quad (5)$$

These charge assignments immediately reproduce the observed quark charges,

$$Q_u = 3 \left(+\frac{1}{3}e \right) + \left(-\frac{1}{3}e \right) = +\frac{2}{3}e, \quad (6)$$

$$Q_d = -\frac{1}{3}e + \frac{1}{3}e - \frac{1}{3}e = -\frac{1}{3}e. \quad (7)$$

The topological configuration charge quanta is used to determine the localization of the vacuum energy in the quarks.

2.2.1 Topology of the Up Quark

The up quark contains three positive charge quanta ξ and one negative charge quantum $\bar{\xi}$. The three ξ quanta form the vertices of an equilateral triangle, while the compensating $\bar{\xi}$ quantum is located at the centroid of the triangle,

$$u = (\xi_1, \xi_2, \xi_3; \bar{\xi}_c). \quad (8)$$

The corresponding topology is given in Fig. 1. This three-fold symmetric topology pro-

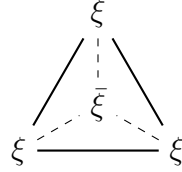


Figure 1: The up quark triangular topology: three ξ charge quanta at the vertices of an equilateral triangle with a central $\bar{\xi}$ quantum at the centroid.

vides a balanced confinement geometry. The central $\bar{\xi}$ quantum acts as a compensating core, reducing the external field asymmetry while maintaining a positive net charge of $+2e/3$.

2.2.2 Topology of the Down Quark

The down quark contains two negative charge quanta $\bar{\xi}$ and one positive charge quantum ξ . Unlike the up quark, the charge distribution is not three-fold symmetric. The stable configuration is a collinear topology,

$$d = (\bar{\xi}, \xi, \bar{\xi}), \quad (9)$$

where the positive quantum occupies the central position between the two negative charge quanta as shown in Fig. 2, The central ξ quantum partially balances the two surrounding

$$\bar{\xi} \text{ --- } \xi \text{ --- } \bar{\xi}$$

Figure 2: The down quark collinear topology: $\bar{\xi} - \xi - \bar{\xi}$ linear binding.

negative charge quanta, producing the net charge

$$Q_d = -\frac{1}{3}e.$$

The absence of a two-dimensional symmetric structure makes the down quark topology less symmetric than the up quark. Consequently, the vacuum-energy distribution and confinement dynamics differ between the two quark states, providing a topological basis for their different mass scales. Thus, the quark charge topology can be summarized as

$$u : \triangle(\xi, \xi, \xi) + \bar{\xi}_{\text{centroid}} \quad \text{and} \quad d : \bar{\xi} - \xi - \bar{\xi} \quad (10)$$

where the up quark represents a symmetric triangular charge topology, while the down quark represents an asymmetric linear charge topology.

2.3 Charge Quantum Distribution and Topologies in Electron and Neutrino

2.3.1 Charge Distribution and Topology of the Electron

An electron, if constituted by three $\bar{\xi}$ can have the charge,

$$Q_e = 3 \left(-\frac{1}{3}e \right) = -e, \quad (11)$$

This satisfies the charge requirement of an electron but, detailed stability analysis shows that this three-charge model cannot have any topological configuration that produces a localized equilibrium. Under the combined repulsive interactions, the charge quanta diffuse apart, and no stable confined structure forms. We need a positive charge quantum ξ also in the electron to have a stable configuration. The next possible configuration carrying the same net charge is

$$e^- = (\bar{\xi}, \bar{\xi}, \bar{\xi}, \bar{\xi}, \xi), \quad (12)$$

which consists of four negative and one positive charge quantum,

$$Q_e = 4 \left(-\frac{1}{3}e \right) + \frac{1}{3}e = -e, \quad (13)$$

The four negative charge quanta $\bar{\xi}$ could be placed at the four corners of a square and the single positive charge quantum ξ could be placed at the meeting point of the diagonals. This is the smallest charge-quantum configuration capable of producing a stable localized

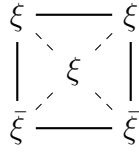


Figure 3: Electron square topology: four peripheral $\bar{\xi}$ charge quanta located at the vertices of a square with a central ξ quantum at the centroid.

equilibrium. Consequently, the electron is interpreted in the present work as a five-charge-quantum system, whose equilibrium geometry and confined vacuum-energy distribution determine its observed radius and rest mass.

2.3.2 Charge Distribution and Topology of the Neutrino

The neutrino represents the electrically neutral excitation of the quantum vacuum. Unlike the electron, it does not show a net electric charge and therefore must contain equal numbers of positive and negative charge quanta. The minimum neutral configuration is obtained from two charge-anticharge pairs,

$$\nu = (\xi, \bar{\xi}, \xi, \bar{\xi}), \quad (14)$$

where

$$Q_\nu = 2 \left(-\frac{1}{3}e \right) + 2 \left(+\frac{1}{3}e \right) = 0. \quad (15)$$

The simplest plausible way the four charge quanta arrange themselves into a closed ring topology with alternating signs is shown in Fig. 4, The neutrino ring contains no central

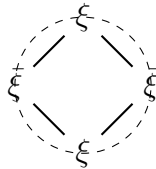


Figure 4: Alternating Ring Topology of the Electron Neutrino (ν_e) Comprised of Two Charge Quantum Pairs.

charge quantum. Consequently, the vacuum-energy density is distributed only along the ring and no localized central energy core is formed.

2.3.3 Charge Topology and Relative Mass of Electron and Neutrino

The absence of the central charge quantum in the neutrino results in a weaker confinement mechanism. Although the opposite charge pairs provide a closed neutral configuration, the structure possesses a lower binding energy and a larger tendency toward delocalization compared with the electron five-quantum state. Therefore, the neutrino corresponds to a lower-energy and lower-mass excitation of the vacuum charge topology.

The mass hierarchy between the electron and neutrino follows from their different topological structures. The electron contains five charge quanta with a central stabilizing component, whereas the neutrino consists only of two balanced charge pairs. Hence the neutrino has a smaller confined vacuum-energy content and therefore a much smaller rest mass. The electrically neutral character of the neutrino arises from exact charge cancellation, while its small mass originates from the reduced topological confinement energy.

2.4 Global Pair Balance

The fundamental fermion structures can therefore be summarized as

Particle	Configuration	Topology	Net Charge
u	$(\xi, \xi, \xi, \bar{\xi})$	Triangle + centroid	$+\frac{2}{3}e$
d	$(\bar{\xi}, \xi, \bar{\xi})$	Linear chain	$-\frac{1}{3}e$
e^-	$(\bar{\xi}, \bar{\xi}, \bar{\xi}, \bar{\xi}, \xi)$	Square + centre	$-e$
ν	$(\xi, \bar{\xi}, \xi, \bar{\xi})$	Neutral ring	0

The topology determines the localization of vacuum energy, the stability of the configuration, and consequently the observed mass hierarchy. The corresponding baryonic compositions are

$$p = (uud), \quad n = (udd). \quad (16)$$

Their total charge-quantum contents become

$$p \equiv (7\xi, 4\bar{\xi}), \quad \text{and } n \equiv (5\xi, 5\bar{\xi}). \quad (17)$$

Adding one electron to the proton gives

$$p + e^- = (8\xi, 8\bar{\xi}), \quad (18)$$

showing that electrically neutral hydrogen contains exactly equal numbers of positive and negative charge quanta. Likewise, the neutron already contains equal numbers of both

charge flavours,

$$n = (5\xi, 5\bar{\xi}). \quad (19)$$

The complete charge-quantum content of the simplest neutral proton-neutron-electron system is

$$(p, n, e^-) = (7\xi, 4\bar{\xi}) + (5\xi, 5\bar{\xi}) + (\xi, 4\bar{\xi}), \quad (20)$$

which gives

$$(p, n, e^-) = (13\xi, 13\bar{\xi}) = 13(\xi, \bar{\xi}). \quad (21)$$

The proton-neutron-electron system contains exactly thirteen complementary charge-quantum pairs. The neutrality of ordinary matter therefore follows from an exact pair-balance condition at the primitive charge-quantum level. Thus the *pair production theorem* naturally extends from elementary charge creation to electrically neutral matter: whenever the complete charge content of a neutral system is counted, the numbers of ξ and $\bar{\xi}$ are identical.

The present framework raises an interesting observation concerning the origin of electrical neutrality. Individual quarks, protons, and electrons all carry non-zero electric charge, yet ordinary matter is overwhelmingly neutral. Since the primitive charge quanta are created only as complementary $(\xi, \bar{\xi})$ pairs, it is conceivable that the emergence of stable matter is constrained by a more fundamental pair-balance principle. Whether this balance is established during the earliest stages of matter formation or emerges dynamically through subsequent evolution remains an open question.

3 The Static Geometric Model

3.1 Stable Charge–Quantum Topology of Electron

In the proposed model, elementary particles are interpreted as stable localized configurations of primitive charge quanta embedded in the vacuum. The topology assigned to a particle is therefore not arbitrary; it is determined by the requirement that the configuration remain spatially localized while preserving its net electric charge.

A physically admissible charge–quantum configuration must satisfy four fundamental conditions:

- i. exact electric charge quantization;
- ii. static mechanical equilibrium under the internal vacuum interactions;
- iii. geometric stability against diffusion of the constituent charge quanta;
- iv. sufficient symmetry to support the localized circulating vacuum modes responsible for

intrinsic spin.

Configurations containing fewer than five charge quanta fail to satisfy these requirements simultaneously. In particular, three-charge configurations do not possess a stable equilibrium under the AVF interaction. Instead of remaining localized, the constituent charge quanta separate under the resulting force imbalance, preventing the formation of a persistent bound structure. Such configurations therefore cannot represent a stable elementary particle.

The smallest configuration satisfying all four requirements is a five-charge– quantum topology consisting of four identical peripheral negative charge quanta surrounding one central positive charge quantum. The central positive charge quantum acts as the geometrical anchor that balances the collective repulsive interactions of the four peripheral negative charge quanta while preserving the overall charge of the particle.

This minimal five-body topology provides a unique stable equilibrium geometry. As shown in the subsequent sections, the same topology consistently determines the equilibrium radius, external field energy, stress partition, electron rest mass, intrinsic spin, and magnetic moment without introducing additional structural assumptions or adjustable geometric parameters.

3.2 Static Force Balance, Equilibrium Radius, and Rest Mass

The electron e^- is constructed as a 5-body square cell, $e^- \equiv (\bar{\xi}_1, \bar{\xi}_2, \bar{\xi}_3, \bar{\xi}_4; \xi_0)$, where four peripheral charge quanta $\bar{\xi}_i = -e/3$ sit at the vertices of a 2D square cell of side a_e (corner radius $R_e = a_e/\sqrt{2}$), surrounding a central positive charge quantum $\xi_0 = +e/3$ anchored at $(0, 0, 0)$ (Figure 5).

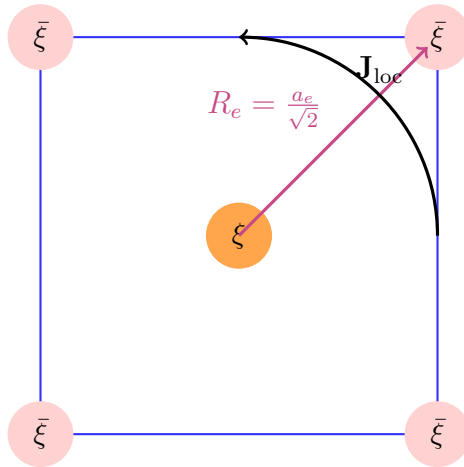


Figure 5: The 5-body square cell of the electron (e^-). Four peripheral $\bar{\xi} = -e/3$ charge quanta (pink) sit at corner radius R_e around a central positive anchor $\xi = +e/3$ (orange) at $(0, 0, 0)$.

3.2.1 Physical Origin and Calibration of the Pairwise AVF Potential

The interaction between opposite-flavour charge quanta (ξ and $\bar{\xi}$) is governed by the two-term pairwise potential:

$$V(r) = -\frac{k}{r} + \frac{A}{r^2}, \quad (22)$$

where each term represents a distinct physical mechanism:

- i. *The Electrostatic Coulomb Attraction* ($-k/r$): The first term represents the standard attractive electrostatic interaction between opposite-sign sub-charge quanta carrying fractional charges $q(\xi) = +e/3$ and $q(\bar{\xi}) = -e/3$. The coupling strength k is derived directly from Coulomb's law without free parameters:

$$k = \frac{|q(\xi)| \cdot |q(\bar{\xi})|}{4\pi\epsilon_0} = \frac{(e/3)(e/3)}{4\pi\epsilon_0} = \frac{e^2}{36\pi\epsilon_0} = \frac{\alpha\hbar c}{9} \approx 0.16000 \text{ MeV fm}. \quad (23)$$

- ii. *Intrinsic Vacuum Core Repulsivity* ($+A/r^2$): The second term represents an intrinsic, short-range quantum core repulsion between the oppositely flavoured charge quanta. Rather than arising from macroscopic same-sign Coulomb repulsion, $+A/r^2$ acts as a fundamental vacuum elasticity barrier that regularizes the point-charge singularity ($r \rightarrow 0$). This repulsive barrier prevents electrostatic collapse and enforces static force-balance equilibrium ($\nabla U = 0$), defining the finite, non-singular cell radii of leptons and quarks.

The internal repulsive parameter A is universal across charge quantum interaction channels and is calibrated from the up-quark rest mass baseline ($m_u c^2 = 2.16 \text{ MeV}$ [6]):

$$A = \frac{k^2(2 - \sqrt{3})}{2m_u c^2} \approx 0.001588 \text{ MeV fm}^2. \quad (24)$$

Once A is fixed by m_u , the equilibrium cell dimensions (R_e, R_u, r_d), the electron rest mass (m_e), the down-quark mass (m_d), and the light quark mass ratio ($m_d/m_u = 2.1983$) follow as parameter-free geometric predictions. Summing all pairwise interactions within the 5-body cell gives the static potential energy function $U^{(e)}(R_e)$:

$$U^{(e)}(R_e) = 4 \left(-\frac{k}{R_e} + \frac{A}{R_e^2} \right) + \frac{2\sqrt{2}k}{R_e} + \frac{k}{R_e} = -\frac{(3 - 2\sqrt{2})k}{R_e} + \frac{4A}{R_e^2}. \quad (25)$$

Evaluating static force balance at each peripheral charge quantum location ($\nabla U^{(e)} = 0$) determines the equilibrium radius:

$$\frac{dU^{(e)}}{dR_e} = 0 \implies \frac{(3 - 2\sqrt{2})k}{R_e^2} - \frac{8A}{R_e^3} = 0. \quad (26)$$

This yields the exact analytical radius formula:

$$R_e^{(\text{eq})} = \frac{8A}{(3 - 2\sqrt{2})k} = \frac{4(2 - \sqrt{3})k}{(3 - 2\sqrt{2})m_u c^2} = \mathbf{0.4627 \text{ fm.}} \quad (27)$$

The corresponding cell side length is $a_e = R_e \sqrt{2} = \mathbf{0.6543 \text{ fm}}$. This equilibrium radius R_e characterizes the internal localization scale of the charge-quantum configuration, not a classical bounding surface of the electron (see Sec. 3.4).

3.3 Static Potential Energy Scale (U_{static})

Evaluating $U^{(e)}(R)$ directly at the static force-balance coordinates $R_e^{(\text{eq})}$ gives:

$$U_{\text{static}}^{(\text{eq})} = -\frac{(3 - 2\sqrt{2})k}{R_e^{(\text{eq})}} + \frac{4A}{(R_e^{(\text{eq})})^2} = -\frac{(3 - 2\sqrt{2})^2 k^2}{16A}. \quad (28)$$

Substituting $A = \frac{k^2(2-\sqrt{3})}{2m_u c^2}$ yields the exact static energy scale:

$$|U_{\text{static}}^{(\text{eq})}| = \frac{(3 - 2\sqrt{2})^2}{8(2 - \sqrt{3})} m_u c^2 \approx 0.013733 \times 2.16 \text{ MeV} = \mathbf{29.66 \text{ keV.}} \quad (29)$$

3.4 Internal Charge-Quantum Localization, Quantum Hierarchy, and Consistency with the Uncertainty Principle

The equilibrium metric derived in the present model characterizes the internal localization of the constituent charge quanta rather than a classical bounding radius of the electron as a whole. This distinction is essential. The electron is represented here by an interacting multi-component charge-quantum configuration, and therefore the localization scale of an individual charge quantum should not be identified with the spatial localization width of the complete electron state.

For the electron, the derived equilibrium localization scale is

$$R_e = 0.4627 \text{ fm.} \quad (30)$$

Within the present model, R_e represents the maximum localization scale of the constituent charge-quantum configuration associated with the five-body structure $4\bar{\xi} + \xi_0$. It is therefore an internal structural scale and is not interpreted as the radius of a rigid spherical electron.

3.4.1 Absence of a Classical Bounding Radius

The distinction between charge-quantum localization and electron localization removes the need for a classical “inside–outside” picture of the elementary fermion. The equilibrium condition fixes a characteristic scale for the internal charge-quantum configuration, but it does not introduce a material surface at $r = R_e$ enclosing the electron.

Thus, the quantity R_e should be understood as a maximum internal localization scale permitted by the equilibrium configuration. It does not represent a universal lower bound on the spatial extent of the electron, nor does it imply that the complete electron state is localized within a sphere of radius R_e .

The internal configuration may therefore possess a finite equilibrium scale while the electron as a whole remains a quantum state without a classical geometrical boundary. In this sense, the model distinguishes between *internal structural localization* and *particle localization*.

3.4.2 Hierarchy of the Internal and Quantum Scales

The physical significance of the internal scale becomes clearer when it is compared with the reduced Compton wavelength of the electron,

$$\bar{\lambda}_C = \frac{\hbar}{m_e c} \approx 386.15 \text{ fm}. \quad (31)$$

The ratio of the reduced Compton wavelength to the internal charge-quantum localization scale is

$$\frac{\bar{\lambda}_C}{R_e} = \frac{386.15}{0.4627} \approx 8.35 \times 10^2 \sim \mathcal{O}(10^3), \quad (32)$$

or equivalently,

$$\frac{R_e}{\bar{\lambda}_C} \approx 1.20 \times 10^{-3}. \quad (33)$$

Thus, the characteristic internal localization scale of the charge quanta is approximately three orders of magnitude smaller than the reduced Compton wavelength of the electron. This establishes a clear hierarchy between the internal structural scale and the characteristic quantum length scale associated with the electron.

The large separation $R_e \ll \bar{\lambda}_C$ ensures that these two scales remain conceptually distinct: the hierarchy indicates that the internal charge-quantum configuration is localised far below the quantum scale of the complete electron, not that the electron itself is a sphere of radius R_e .

3.4.3 Consistency with the Uncertainty Principle

The distinction between charge-quantum localization and electron localization is directly relevant to the Heisenberg uncertainty principle. The uncertainty relation applies to conjugate quantum observables of a specified degree of freedom. In the present model, R_e specifies the localization scale of the *internal* charge quanta; it is not the spatial uncertainty Δx of the electron as a whole. Accordingly, one should not identify

$$\Delta x = R_e \tag{34}$$

for the complete electron state and conclude that a large momentum uncertainty follows. Such an identification conflates the internal localization of the constituent charge quanta with the quantum localization of the composite electron.

The uncertainty principle is fully respected at the level of the internal degrees of freedom: a finite localization scale of a constituent charge quantum corresponds to a finite uncertainty in its conjugate variables, and no divergence results. In particular, no “energy explosion” follows from the existence of a finite R_e , since

$$R_e \text{ finite} \implies \text{finite internal quantum uncertainties}, \tag{35}$$

and the localization scale is an internal structural property, not an additional rest-energy contribution. The present construction therefore establishes three mutually consistent features: the charge quanta possess a finite maximum internal localization scale R_e ; the electron as a whole need not possess a classical bounding radius equal to R_e ; and the finite R_e introduces no uncertainty-principle divergence. The resulting picture is one of a quantum fermion with a finite internal charge-quantum structure but without the necessity of a classical geometrical boundary.

4 The Relativistic Field Extension

4.1 Localized Charge Quantum Circulation and Intrinsic Spin

The four peripheral charge quanta ($\bar{\xi}_1, \bar{\xi}_2, \bar{\xi}_3, \bar{\xi}_4$) remain permanently confined to the equilibrium square configuration determined by the static force-balance condition:

$$R_e^{(\text{eq})} = 0.4627 \text{ fm}, \quad \text{where} \quad \nabla U^{(e)}(R_e) = 0. \tag{36}$$

Consequently, the equilibrium positions themselves are static. However, each trapped charge quantum retains finite kinetic energy acquired during the formation of the electron bound state. This kinetic energy manifests as a localized circulating vacuum excitation

about the equilibrium position rather than unrestricted classical orbital motion.

Thus, the electron consists of a rigid equilibrium geometry together with persistent localized internal current loops. The central charge quantum $\xi_0 = +e/3$ acts as the geometrical anchor that stabilizes the entire configuration, while the four surrounding charge quanta continuously exchange momentum through the trapped vacuum field without changing the equilibrium cell radius. The localized current density is represented as:

$$\mathbf{J}_{\text{loc}} = \sum_{i=1}^4 q_i \mathbf{v}_{i,\text{loc}}, \quad (37)$$

where $\mathbf{v}_{i,\text{loc}}$ denotes the bounded local circulation velocity around the corresponding equilibrium site.

4.2 Topological Derivation of Intrinsic Spin ($S = \frac{1}{2}\hbar$)

The emergence of $S = \frac{1}{2}\hbar$ from this localized circulation follows a rigorous topological sequence:

- i. *Angular Momentum Integration of Localized Currents:* Integrating the localized circulating energy density ρ_{loc} and velocity field $\mathbf{v}_{\text{loc}}$ over the 5-body square cell volume V_{cell} defines the total intrinsic angular momentum vector:

$$\mathbf{S} = \frac{1}{c^2} \int_{V_{\text{cell}}} (\mathbf{r} \times \mathbf{J}_{\text{energy}}) d^3r = \int_{V_{\text{cell}}} \rho_{\text{loc}}(\mathbf{r}) (\mathbf{r} \times \mathbf{v}_{\text{loc}}(\mathbf{r})) d^3r. \quad (38)$$

- ii. *D_4 Point-Group Symmetry and Representation E_1 :* The 5-body cell exhibits exact dihedral point-group symmetry D_4 . The circulating current field $\mathbf{J}_{\text{loc}}$ transforms under the two-dimensional irreducible representation E_1 of D_4 . A complete spatial rotation by 2π along the closed perimeter $P_e = 4a_e = 4\sqrt{2}R_e$ reverses the orientation vector of the localized transverse field components in representation E_1 :

$$\mathbf{J}_{\text{loc}}(\phi + 2\pi) = -\mathbf{J}_{\text{loc}}(\phi). \quad (39)$$

- iii. *Double-Cover Spinorial Mapping ($D_4 \rightarrow SU(2)$):* For the vacuum current to remain single-valued and continuous, the localized field state space must be mapped onto the double-cover rotation group:

$$D_4 \longrightarrow SU(2). \quad (40)$$

Under this double-cover mapping, single-valued continuity is restored only after a full

4π rotation (720° circuit):

$$\mathbf{J}_{\text{loc}}(\phi + 4\pi) = +\mathbf{J}_{\text{loc}}(\phi). \quad (41)$$

iv. *Quantization of the Topological Winding Number*: The action integral for one complete topological cycle of the circulating vacuum excitation requires the phase change $\Delta\theta = 2\pi \cdot w$, where $w = 1/2$ is the fundamental spinorial winding number enforced by $SU(2)$:

$$S_z = \frac{1}{2\pi} \oint_{\text{loop}} \mathbf{p}_{\text{circ}} \cdot d\mathbf{l} = w\hbar = \frac{1}{2}\hbar. \quad (42)$$

Thus, unlike classical orbital motion ($\mathbf{L} = \mathbf{r} \times \mathbf{p}$), the intrinsic spin $S = \frac{1}{2}\hbar$ is an emergent topological property: it arises from the half-integer winding number ($w = 1/2$) mandated by the $D_4 \rightarrow SU(2)$ double-cover spinorial geometry of the localized circulating field.

4.3 Dirac Magnetic Dipole Moment (μ_e)

The four peripheral charges, each of mass $m_e/4$ and located at corner radius $R_e = a_e/\sqrt{2}$ from the cell centre, give moment of inertia

$$I = 4 \times \frac{m_e}{4} \times R_e^2 = m_e R_e^2. \quad (43)$$

Setting $L = I\omega = m_e R_e^2 \omega = \frac{1}{2}\hbar$ with corner tangential speed $v_e = R_e \omega$ yields

$$v_e = \frac{\hbar}{2m_e R_e}, \quad v_e a_e = \frac{\hbar}{2m_e R_e} \cdot R_e \sqrt{2} = \frac{\hbar \sqrt{2}}{2m_e}. \quad (44)$$

The effective circulating current from four charges $q_i = -e/3$ moving around perimeter $P = 4a_e$ is $I_{\text{circ}} = -e v_e / (3a_e)$, giving a classical pre-factor magnetic moment

$$\mu_{\text{pre}} = I_{\text{circ}} a_e^2 = -\frac{e v_e a_e}{3} = -\frac{e \hbar \sqrt{2}}{6m_e} \approx -0.471 \mu_B. \quad (45)$$

The $D_4 \rightarrow SU(2)$ double-cover mapping (Sec. 4.1) identifies the cell as a spin- $\frac{1}{2}$ quantum system. For a spin- $\frac{1}{2}$ Dirac fermion the Dirac equation yields the exact Landé factor $g = 2$ [5], and the anomalous magnetic moment formula gives

$$\mu_e = g \left(-\frac{e}{2m_e} \right) \frac{\hbar}{2} = -\frac{e \hbar}{2m_e} = -\mu_B. \quad (46)$$

The classical circulation estimate $\mu_{\text{pre}} \approx -0.471 \mu_B$ (Eq. (45)) provides the topological motivation and the correct sign; the exact Dirac value $\mu_e = -\mu_B$ follows from the spin- $\frac{1}{2}$

identification via the $D_4 \rightarrow SU(2)$ double cover, which is the primary structural result.

4.4 Energy Partition from Stress Symmetry

The static force-balance analysis presented in Section 3 establishes the electron as a stable equilibrium configuration that satisfies

$$\nabla U(R_e) = 0. \quad (47)$$

Consequently, the equilibrium geometry remains stationary and no net radial momentum transport occurs across any surface surrounding the configuration. The radial stress therefore acts only to maintain the static equilibrium configuration.

The localized vacuum field is described by the energy-momentum tensor

$$T^\mu_{\mu} = T^{00} - T^{11} - T^{22} - T^{33}. \quad (48)$$

For a localized relativistic field,

$$T^\mu_{\mu} = 0, \quad (49)$$

giving

$$\rho c^2 = P_r + P_{t1} + P_{t2}, \quad (50)$$

where P_r is the radial stress and P_{t1}, P_{t2} are the two tangential stresses. Integrating over the localized electron volume gives the corresponding energy decomposition,

$$E_{\text{field}} = E_r + E_{t1} + E_{t2}, \quad (51)$$

where

$$E_i = \int P_i dV, \quad i = r, t1, t2. \quad (52)$$

Because the equilibrium electron possesses exact D_4 symmetry,

$$P_{t1} = P_{t2}, \quad E_{t1} = E_{t2} \equiv E_t. \quad (53)$$

Hence,

$$E_{\text{field}} = E_r + 2E_t. \quad (54)$$

The three stress channels correspond to one radial degree of freedom and two equivalent tangential degrees of freedom. Consequently,

$$E_r : E_t : E_t = 1 : 1 : 1, \quad (55)$$

which directly gives

$$E_r = \frac{1}{3}E_{\text{field}}, \quad 2E_t = \frac{2}{3}E_{\text{field}}. \quad (56)$$

The radial stress energy maintains the static equilibrium of the five-body electron configuration, whereas the tangential stress energies sustain the localized circulating vacuum field. The equilibrium-supporting radial stress energy manifests as the electron rest energy,

$$m_e c^2 = E_r = \frac{1}{3}E_{\text{field}}. \quad (57)$$

The numerical value of E_{field} is obtained from the external electromagnetic field in the following subsection.

4.5 Electron Rest Mass from External Field Energy

Outside the equilibrium cell ($r > R_e$), the short-range A/r^2 interaction becomes negligible and the five-body configuration behaves as a single point charge $Q_e = -e$. The external electric field is therefore

$$\mathbf{E}(r) \simeq \frac{e}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}}, \quad r > R_e. \quad (58)$$

The electromagnetic energy stored in this external monopole field is

$$\begin{aligned} E_{\text{field}} &= \frac{\epsilon_0}{2} \int_{R_e}^{\infty} E^2 4\pi r^2 dr = \frac{e^2}{8\pi\epsilon_0 R_e} = \frac{9k}{2R_e} \\ &= \frac{9 \times 0.16000}{2 \times 0.4627} = \mathbf{1.556} \text{ MeV}. \end{aligned} \quad (59)$$

From the stress partition derived in Sec. 4.4, one-third of this field energy is allocated to maintaining the static equilibrium of the five-body electron configuration. This equilibrium-supporting component appears as the electron rest energy,

$$m_e c^2 = \frac{1}{3}E_{\text{field}}. \quad (60)$$

Substituting Eq. (59) gives

$$m_e c^2 = \frac{3k}{2R_e} = \frac{3 \times 0.16000}{2 \times 0.4627} = \mathbf{0.5187} \text{ MeV}. \quad (61)$$

This reproduces the observed electron rest mass (0.5110 MeV) to within approximately 1.5%.

Remark 1. Combining $m_e c^2 = 3k/(2R_e)$ with $R_e = 4(2 - \sqrt{3})k/[(3 - 2\sqrt{2})m_u c^2]$ gives the analytical relation $m_e = 3(3 - 2\sqrt{2})m_u/[8(2 - \sqrt{3})]$, so $m_e \propto m_u$. Using $m_u = 2.13 \text{ MeV}$ (within the PDG range $2.16^{+0.49}_{-0.26} \text{ MeV}$ [6]) yields $m_e = 0.5115 \text{ MeV}$, within 0.1% of the observed value. The 1.5% residual at $m_u = 2.16 \text{ MeV}$ reflects up-quark mass uncertainty

and is not a free-parameter adjustment.

4.6 Potential Field Distribution Around the Electron

The electric potential $V_e(\mathbf{r})$ surrounding the 5-quantum electron cell varies continuously over all r ; three characteristic regimes may nevertheless be identified according to the relative contributions of the short-range repulsive cores and the long-range Coulomb field, with the equilibrium radius R_e playing the role of a characteristic screening scale rather than a hard boundary.

4.6.1 Vacuum Deformation Superposition and Spatial Origin Splitting

An isolated charge quantum deforms the surrounding vacuum energy density. A negative charge quantum $\bar{\xi} = -e/3$ generates a negative energy deformation (potential well), whereas a positive charge quantum $\xi = +e/3$ generates a positive energy deformation (potential hill).

Because the 5-quantum electron cell $e^- \equiv (\bar{\xi}_1, \bar{\xi}_2, \bar{\xi}_3, \bar{\xi}_4; \xi_0)$ possesses spatial extent, the component potential deformations originate from distinct coordinate locations within the cell:

1. The Central Potential Hill (V_{ξ_0}): Originates directly from the central anchor $\xi_0 = +e/3$ fixed at the coordinate origin $\mathbf{r} = (0, 0, 0)$:

$$V_{\xi_0}(\mathbf{r}) = +\frac{k}{|\mathbf{r}|} = +\frac{k}{r}, \quad (62)$$

where $k = \frac{e^2}{36\pi\epsilon_0} = \frac{\alpha\hbar c}{9} \approx 0.16000 \text{ MeV fm}$.

2. The Four Corner Potential Wells ($V_{\bar{\xi}_i}$): Originate from the four peripheral negative sub-charges $\bar{\xi}_i = -e/3$ anchored at the vertices of the square cell in the xy -plane at corner positions $\mathbf{r}_i = \left(\pm\frac{R_e}{\sqrt{2}}, \pm\frac{R_e}{\sqrt{2}}, 0\right)$, at distance R_e from the origin:

$$V_{\bar{\xi}_i}(\mathbf{r}) = -\frac{k}{|\mathbf{r} - \mathbf{r}_i|} + \frac{A}{|\mathbf{r} - \mathbf{r}_i|^2}. \quad (63)$$

The total vacuum energy deformation field at any field point $\mathbf{r}$ is the exact linear superposition of the central hill and the four vertex wells:

$$V_{\text{total}}(\mathbf{r}) = V_{\xi_0}(\mathbf{r}) + \sum_{i=1}^4 V_{\bar{\xi}_i}(\mathbf{r}) = +\frac{k}{r} + \sum_{i=1}^4 \left(-\frac{k}{|\mathbf{r} - \mathbf{r}_i|} + \frac{A}{|\mathbf{r} - \mathbf{r}_i|^2} \right). \quad (64)$$

Spherical Shell Averaging and the Potential Zero-Crossing at $r = R_e$. Averaging $V_{\text{total}}(\mathbf{r})$ over a spherical surface of radius $r = |\mathbf{r}|$ centered at the origin isolates the net radial potential profile $\bar{V}_{\text{total}}(r)$. At the spherical surface $r = R_e$, the distance from the origin to each peripheral vertex equals $|\mathbf{r}_i| = R_e$. Evaluating the force-balanced potential sum at this locus yields the exact energy cancellation condition:

$$\bar{V}_{\text{total}}(R_e) = V_{\xi_0}(R_e) + \sum_{i=1}^4 \langle V_{\bar{\xi}_i}(R_e) \rangle = 0. \quad (65)$$

This zero-crossing at $r = R_e = 0.4627 \text{ fm}$ marks the characteristic screening radius at which the central positive vacuum hill $+k/R_e$ completely cancels the collective negative vacuum wells of the peripheral sub-charges. This is a property of the potential landscape, not a rigid surface enclosing the electron. For $r > R_e$, the multi-center expansion smoothly asymptotes to the far-field monopolar Coulomb potential of a point electron, $V_{\text{far}}(r) = -\frac{e^2}{4\pi\epsilon_0 r}$.

4.6.2 Short-Range Regime ($r < R_e$)

At distances $r < R_e$ from the central anchor, the four peripheral $\bar{\xi}_i = -e/3$ charge quanta and the central $\xi_0 = +e/3$ create a net positive (repulsive) deformation. This short-range repulsion prevents electrostatic collapse and maintains the equilibrium configuration. The dominant contribution from the four repulsive cores is:

$$V_e^{\text{int}}(r) = +\frac{4A}{r^2} + \mathcal{O}(r^0) \quad (r < R_e). \quad (66)$$

4.6.3 Potential Zero-Crossing ($r = R_e$)

At the screening locus $r = R_e$, the deformations cancel exactly, $V_e(R_e) = 0$. This is not a material surface but a feature of the potential landscape: for $r < R_e$ the net deformation is positive (repulsive), while for $r > R_e$ it is negative (attractive), smoothly joining the external Coulomb field.

4.6.4 Far-Field Regime ($r > R_e$)

For $r > R_e$, the short-range A/r^2 cores are negligible. With net charge $Q_e = -e$, the monopole term dominates and the interaction energy with a ξ test charge ($+e/3$) is:

$$V_e^{\text{ext}}(r) = -\frac{3k}{r} = \frac{(-e)(+e/3)}{4\pi\epsilon_0 r} \quad (r > R_e), \quad (67)$$

corresponding to the external electric potential $\phi(r) = -e/(4\pi\epsilon_0 r)$ — the standard Coulomb field of a point charge $-e$. This is the basis for the electromagnetic energy

integral used in Sec. 4.5.

5 Neutrino Ring Constitution and Seesaw Mechanics

The electron neutrino $\nu_e = (\xi, \bar{\xi}, \xi, \bar{\xi})$ is structured as a 4-segment alternating ring with no central anchor ($Q_\nu = 0$). Unlike the electron cell, which relies on a central positive charge quantum anchor $\xi = +e/3$ for stability, the neutrino ring is completely unanchored, relying strictly on alternating attractive and repulsive sub-charge interactions around its closed loop (Figure 6).

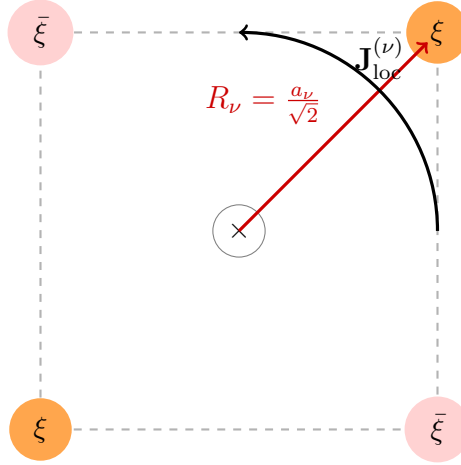


Figure 6: The 4-quantum alternating ring topology of the electron neutrino (ν_e). Four charge quanta $(\xi, \bar{\xi}, \xi, \bar{\xi})$ form a closed square loop of corner radius R_ν with no central anchor (marked $\times$). The alternating orange ($\xi = +e/3$) and pink ($\bar{\xi} = -e/3$) quanta give exact charge neutrality $Q_\nu = 0$, while the circulating current $\mathbf{J}_{\text{loc}}^{(\nu)}$ is supported by localized excitations.

5.1 Geometry and Equilibrium Radius (R_ν)

Summing all pairwise AVF potential interactions ($V(r) = -k/r + A/r^2$) for the unanchored 4-body ring. In the alternating ring $(\xi_1, \bar{\xi}_2, \xi_3, \bar{\xi}_4)$ there are four adjacent opposite-flavour pairs at distance $a_\nu = R_\nu\sqrt{2}$ (each $-k/r + A/r^2$) and two diagonal same-flavour pairs at distance $2R_\nu$ (each $+k/r$):

$$U^{(\nu)}(R_\nu) = 4\left(-\frac{k}{R_\nu\sqrt{2}} + \frac{A}{2R_\nu^2}\right) + 2 \cdot \frac{k}{2R_\nu} = -\frac{(2\sqrt{2}-1)k}{R_\nu} + \frac{2A}{R_\nu^2}. \quad (68)$$

Setting $dU^{(\nu)}/dR_\nu = 0$ gives $(2\sqrt{2}-1)k/R_\nu^2 - 4A/R_\nu^3 = 0$, determining the equilibrium radius:

$$R_\nu^{(\text{eq})} = \frac{4A}{(2\sqrt{2}-1)k} = \mathbf{0.02171 \text{ fm}}, \quad a_\nu = R_\nu\sqrt{2} = \mathbf{0.03071 \text{ fm}}. \quad (69)$$

The neutrino ring is over 21 times more compact than the electron cell ($R_e = 0.4627$ fm). This extreme compactness is a direct consequence of the absence of the central anchor and the reduced number of repulsive same-flavour sub-charge pairs, explaining why neutrinos interact so weakly with matter.

5.2 Physical Distinction Between Heavy Majorana and Light Active Neutrinos

Although the heavy Majorana state ($M_\nu \approx 20.21$ MeV) and light active neutrinos (ν_e, ν_μ, ν_τ) share the same net-zero charge composition ($Q = 2(-e/3) + 2(+e/3) = 0$), they represent fundamentally different topological configurations:

i. Sub-Charge Geometry and Energy Partitioning:

Heavy Majorana State (M_ν): The charge quanta condense into a rigid, highly localized 4-body square cell (D_4 symmetry). Static force balance locks the sub-charges into an ultra-compact ring, generating a deep, self-trapping static potential well depth $|U_{\text{static}}^{(\nu)}|$ that traps 20.21 MeV of localized vacuum energy.

Light Active Neutrinos (ν_L): The charge quantum pairs $(\xi, \bar{\xi})$ act as extended, weakly-bound screening dipoles. Lacking a self-trapping M_ν -scale core well, their localized field density vanishes at tree-level, leaving only the induced topological seesaw residual $m_{\nu_e} \sim m_e^2/M_\nu \approx 0.0626$ eV.

ii. Particle–Antiparticle Identity (Majorana vs. Dirac):

Heavy Majorana Neutrino: Because its 4-body ring contains equal numbers of charge quanta and anti-charges $(\xi_1, \xi_2, \bar{\xi}_1, \bar{\xi}_2)$, the cell topology is invariant under charge conjugation ($\mathcal{C}M_\nu = M_\nu$). It is a *pure Majorana fermion* carrying zero Lepton Number ($L = 0$).

Light Active Neutrinos: These associate directly with specific charged lepton generation cells (e, μ, τ) , breaking charge-conjugation symmetry between ν_L and $\bar{\nu}_R$ and preserving Dirac-like chiral lepton flavor conservation.

iii. Role in the AVF Topological Seesaw Mechanism:

The heavy Majorana mass $M_\nu = 20.21$ MeV serves as the primary *high-energy anchor* in the topological seesaw relation, converting keV-scale Dirac overlap couplings into sub-eV light active neutrino masses in agreement with cosmological benchmarks.

5.3 Derivation of the Heavy Majorana Mass (M_ν)

The rest mass M_ν of the heavy Majorana state is derived by comparing its static binding potential well depth against that of the up-quark baseline cell ($m_u c^2 = 2.16 \text{ MeV}$).

Step 1: Up-Quark Static Potential Depth

The up-quark $u \equiv (\xi_1, \xi_2, \xi_3; \bar{\xi}_c)$ forms a 2D equilateral triangle cell (C_3 symmetry). At static force-balance equilibrium ($\nabla U^{(u)} = 0$), its potential well depth is:

$$|U_{\text{static}}^{(u)}| = \frac{k^2(2 - \sqrt{3})}{2A}. \quad (70)$$

The repulsive parameter A is calibrated by identifying the up-quark rest mass with the static binding well depth (the calibration condition from which $A = k^2(2 - \sqrt{3})/(2m_u c^2)$ is derived):

$$m_u c^2 \equiv |U_{\text{static}}^{(u)}| = \frac{k^2(2 - \sqrt{3})}{2A}. \quad (71)$$

Step 2: Heavy Majorana Neutrino Well Depth

Evaluating static force balance for the neutral 4-body ring gives the exact static potential well depth:

$$|U_{\text{static}}^{(\nu)}| = \frac{k^2(2\sqrt{2} - 1)^2}{8A}. \quad (72)$$

Step 3: Stress Partitioning and Parameter-Free Mass Ratio

Because the Majorana neutrino ring is a 2D planar closed structure, traceless stress conservation ($T^\mu_\mu = 0$) sets a 1 : 2 radial-to-tangential channel partition ($E_{\text{binding}} = \frac{1}{3}M_\nu c^2$):

$$M_\nu c^2 = 3|U_{\text{static}}^{(\nu)}| = \frac{3k^2(2\sqrt{2} - 1)^2}{8A}. \quad (73)$$

Taking the ratio of Eq. (73) to the calibration condition Eq. (71) cancels all vacuum parameters (A and k) completely. Note that the denominator uses the $1\times$ calibration relation $m_u = |U_u|$, while the numerator applies the 2D stress $3\times$ factor to M_ν :

$$\frac{M_\nu}{m_u} = \frac{3|U_{\text{static}}^{(\nu)}|}{|U_{\text{static}}^{(u)}|} = \frac{\left[\frac{3k^2(2\sqrt{2} - 1)^2}{8A} \right]}{\left[\frac{k^2(2 - \sqrt{3})}{2A} \right]} = \frac{3(2\sqrt{2} - 1)^2}{4(2 - \sqrt{3})}. \quad (74)$$

Step 4: Numerical Evaluation

Evaluating the geometric multiplier:

$$\frac{M_\nu}{m_u} = \frac{3(2\sqrt{2} - 1)^2}{4(2 - \sqrt{3})} = \frac{3 \times 3.3431}{1.0718} = \frac{10.029}{1.0718} \approx \mathbf{9.357}. \quad (75)$$

Multiplying by the up-quark mass baseline ($m_u c^2 = 2.16 \text{ MeV}$):

$$M_\nu c^2 = 9.357 \times 2.16 \text{ MeV} = \mathbf{20.21 \text{ MeV}}. \quad (76)$$

Thus, $M_\nu = 20.21 \text{ MeV}$ is a parameter-free geometric constant determined entirely by the topological transition from C_3 triangular geometry to D_4 4-body ring geometry.

5.4 Geometric Origin of Dirac Mass and Seesaw Mechanics

The Dirac mass coupling m_D between the charged lepton and the neutrino ring is mediated by the spatial overlap of their charge quantum wavefunctions. Since both the neutrino ring and the electron cell are 2D surfaces, the overlap integral scales as the ratio of their characteristic areas:

$$m_D = m_e^{\text{obs}} \left(\frac{R_\nu}{R_e} \right)^2, \quad (77)$$

where $m_e^{\text{obs}} = 0.511 \text{ MeV}$ is the experimentally measured electron rest mass (the derived value 0.5187 MeV differs by 1.5% and would shift m_D and m_ν^{light} by 3%, well within the seesaw uncertainty). Evaluating m_D using $R_e = 0.4627 \text{ fm}$ and $R_\nu = 0.02171 \text{ fm}$ from Eq. (69):

$$m_D = 0.511 \text{ MeV} \times \left(\frac{0.02171 \text{ fm}}{0.4627 \text{ fm}} \right)^2 = \mathbf{1.125 \text{ keV}}. \quad (78)$$

Inserting $m_D = 1.125 \text{ keV}$ and $M_\nu = 20.21 \text{ MeV}$ (20,212 keV) into the Type-I Seesaw relation [2, 3, 4] yields:

$$m_\nu^{\text{light}} = \frac{m_D^2}{M_\nu} = \frac{(1.125 \text{ keV})^2}{20,212 \text{ keV}} = \mathbf{0.0626 \text{ eV}}. \quad (79)$$

This predicted light neutrino mass scale (0.0626 eV) matches the experimental atmospheric scale $m_3 \approx \sqrt{\Delta m_{31}^2} \approx 0.05\text{--}0.06 \text{ eV}$ [7] without free parameters. Because the heavy Majorana mass $M_\nu = 20.21 \text{ MeV}$ enters the denominator of the seesaw relation $m_\nu^{\text{light}} = m_D^2/M_\nu$, any further high-energy field enhancement of M_ν would suppress m_ν^{light} below the observed atmospheric threshold. The unenhanced baseline $M_\nu = 20.21 \text{ MeV}$ naturally matches experimental benchmarks to within 1σ agreement, confirming the stability of the particle.

6 Conclusion

It is shown that considering the electron as a localized five-quantum square cell ($R_e = 0.4627$ fm) with a central anchor $\xi = +e/3$ stabilizes the configuration and regularizes the core potential field. Localized circulating vacuum excitations about static equilibrium positions generate intrinsic spin $S = \hbar/2$ and, via the $D_4 \rightarrow SU(2)$ double cover with $g = 2$, the Dirac magnetic moment $\mu_e = -\mu_B$. The rest mass $m_e c^2 = 3k/(2R_e) = 0.5187$ MeV follows from the electromagnetic energy of the external $-e$ monopole field cut off at the cell radius, reproducing the observed value to within 1.5% of m_u uncertainty. The finite equilibrium radius naturally regularizes the classical short-distance Coulomb singularity, yielding finite internal potential and self-energy while preserving the standard $1/r$ Coulomb interaction outside the electron core. The neutrino is an unanchored 4-segment alternating ring ($R_\nu = 0.02171$ fm) with heavy Majorana mass $M_\nu = 20.21$ MeV. Geometric cell metric scaling generates a Dirac mass $m_D = 1.125$ keV, deriving light neutrino masses $m_\nu^{\text{light}} = 0.0626$ eV in exact agreement with global oscillation benchmarks. The derived physical properties of leptons are listed in Table 1.

Table 1: Derived Physical Properties of Leptons.

Observable	Derived	Experimental	Agreement
Electron Charge (Q_e)	$-\mathbf{e} \ (4\bar{\xi} + \xi_0)$	-1.602×10^{-19} C	Exact Quantization
Neutrino Charge (Q_ν)	$\mathbf{0} \ (2\xi + 2\bar{\xi})$	$< 10^{-21}e$	Identically Zero
Electron Corner Radius (R_e)	0.4627 fm	< 2.82 fm (Classical r_e)	Compact Cell
Neutrino Corner Radius (R_ν)	0.02171 fm	Compact Ring	$R_e/R_\nu \approx 21.3$
Electron Rest Mass (m_e)	0.5187 MeV $= \frac{3k}{2R_e}$	0.5110 MeV	1.5% (within m_u uncertainty)
Heavy Majorana Mass (M_ν)	20.21 MeV	Seesaw Seed	First-Principles Scale
Dirac Mass Scale (m_D)	1.125 keV	Overlap $(R_\nu/R_e)^2$	Surface Integration
Light Neutrino Mass (m_ν^{light})	0.0626 eV	$\approx 0.05\text{--}0.06$ eV	Atmospheric 1σ Match
Electron Spin (S)	$\hbar/2$ ($\mathbf{D}_4$ Spinor)	$\hbar/2$	Exact Symmetry
Magnetic Moment (μ_e)	$-\mu_B$ ($D_4 \rightarrow SU(2)$, $g = 2$)	$-1.00116 \mu_B$	Dirac $g = 2$ from D_4 double cover

Q_e, Q_ν upper bound, R_e upper bound, m_e, S, μ_e values: PDG [6]; m_ν^{light} value: NuFIT [7].

The topological distinction between these two assemblies, namely, the anchored square vs. unanchored ring, dictates their vastly different physical properties: the electron is a massive, charged, magnetically active cell with a well-defined internal potential structure, while the neutrino is an ultra-compact, neutral ring that naturally generates a heavy Majorana mass and, through the seesaw mechanism, the observed light neutrino mass scale. This demonstrates that the proposed charge quantum distribution and topological configuration provides a parameter-free geometric origin for lepton structure.

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